

## 1. Agnostic learning

## 1. Risk and best-in-class target

For binary zero-one loss and  $S = ((x_i, y_i))_{i=1}^m \sim D^m$ ,

$$R(h) = P_{(X,Y) \sim D}(h(X) \neq Y), \quad \hat{R}_{S(h)} = \frac{1}{m} \sum_{i=1}^m \mathbf{1}_{\{h(x_i) \neq y_i\}}.$$

$\mathcal{H}$  is **agnostic PAC learnable** if there are one learner  $A$  and a sample complexity  $m_{\mathcal{H}}$  such that, for every  $\varepsilon, \delta \in (0, 1)$ , every joint distribution  $D$ , and every  $m \geq m_{\mathcal{H}}(\varepsilon, \delta)$ ,

$$P\left(R(A(S)) \leq \inf_{h \in \mathcal{H}} R(h) + \varepsilon\right) \geq 1 - \delta.$$

$\varepsilon$  bounds **excess risk** above the best risk in the declared class; it is not distance from zero. The infimum need not be attained: compare with  $h_\rho$  satisfying  $R(h_\rho) \leq \inf_h R(h) + \rho$ , then let  $\rho \rightarrow 0$ .

## 2. Approximate ERM and tolerance roles

$h_S = A(S)$  is an  $\eta$ -approximate ERM output if

$$\hat{R}_{S(h_S)} \leq \inf_{h \in \mathcal{H}} \hat{R}_{S(h)} + \eta.$$

- $\eta$ : empirical optimization error;
- $\rho$ : proof-only population-comparator tolerance;
- $\alpha$ : empirical/population deviation tolerance;
- $\varepsilon$ : requested final excess-risk tolerance.

Do not merge  $\eta$  with  $\rho$ . A successful output on one observed sample is not a PAC proof; the probability guarantee must hold over fresh samples for every allowed  $D$ .

## 3. Uniform convergence and ERM chain

The **uniform convergence** event is

$$E_\alpha = \left\{ \sup_{h \in \mathcal{H}} |\hat{R}_{S(h)} - R(h)| \leq \alpha \right\}.$$

Uniform convergence requires  $P(E_\alpha) \geq 1 - \delta$  for all distributions once  $m$  is sufficiently large. The shared event covers the data-selected  $h_S$  and the comparator  $h_\rho$  simultaneously. On  $E_\alpha$ ,

$$\begin{aligned} R(h_S) &\leq \hat{R}_{S(h_S)} + \alpha \\ &\leq \hat{R}_{S(h_\rho)} + \alpha + \eta \\ &\leq R(h_\rho) + 2\alpha + \eta \\ &\leq \inf_{h \in \mathcal{H}} R(h) + \rho + 2\alpha + \eta. \end{aligned}$$

Letting  $\rho \rightarrow 0$  gives excess risk at most  $2\alpha + \eta$ . For exact ERM,  $\eta = 0$  and the usual choice is  $\alpha = \frac{\varepsilon}{2}$ . Pointwise concentration for each fixed  $h$  does not by itself create this simultaneous event.

## 4. Finite class check

For fixed  $h$  under zero-one loss, Hoeffding gives

$$P\left(|\hat{R}_{S(h)} - R(h)| > \alpha\right) \leq 2 \exp(-2m\alpha^2).$$

If  $\mathcal{H}$  is finite, a union bound makes it sufficient that

$$m \geq \frac{\ln\left(2 \frac{|\mathcal{H}|}{\delta}\right)}{2\alpha^2}.$$

For exact agnostic ERM, substitute  $\alpha = \frac{\varepsilon}{2}$ :

$$m \geq 2 \frac{\ln\left(2 \frac{|\mathcal{H}|}{\delta}\right)}{\varepsilon^2}.$$

This has logarithmic dependence on class size and the agnostic  $\frac{1}{\varepsilon^2}$  scale. It is not the realizable, consistent  $\frac{1}{\varepsilon}$  route.

## 2. Learnability boundary

## 1. Fundamental Theorem

In **distribution-free binary classification with zero-one loss**, i.i.d. samples, suitable measurability, the following are equivalent:

1. uniform convergence;
2. every ERM rule is agnostic PAC;
3. agnostic PAC learnability;
4. realizable PAC learnability;
5. every ERM rule is realizable PAC;
6.  $\text{VCdim}(\mathcal{H}) < \infty$ .

The equivalence is scoped: do not export it unchanged to arbitrary losses or nonmeasurable classes. The ERM chain gives uniform convergence  $\rightarrow$  agnostic ERM success; agnostic learning implies realizable learning by restriction; No-Free-Lunch supplies the necessity of finite VC dimension.

## 2. No-Free-Lunch mechanism

For every binary learner  $A$  and sample size  $m$  with  $2m \leq |\mathcal{X}|$ , there is a realizable distribution with

$$P\left(R(A(S)) \geq \frac{1}{8}\right) \geq \frac{1}{7}.$$

For an infinite-VC class, choose a shattered support  $C = \{x_1, \dots, x_{2m}\}$ . A sample of size  $m$  sees at most  $m$  distinct points, so unseen points carry at least half the probability mass. Shattering realizes every target labeling; under random target bits, each unseen label remains a fair bit and any prediction is wrong with conditional probability  $\frac{1}{2}$ . Averaging identifies one fixed realizable hard target.

**Quantifiers:** for every learner, there exists a hard problem. Infinite class cardinality alone does not imply failure; the relevant richness is infinite VC dimension.

## 3. Random-sign complexity

## 1. Loss class and Rademacher complexity

Let  $\mathcal{Z} = \mathcal{X} \times \{0, 1\}$  and

$$g_{h(x,y)} = \mathbf{1}_{\{h(x) \neq y\}}, \quad \mathcal{G} = \{g_h : h \in \mathcal{H}\}.$$

For fixed  $S = (z_1, \dots, z_m)$ , the **empirical Rademacher complexity** is

$$\hat{\mathfrak{R}}_{S(\mathcal{G})} = E_\sigma \left[ \sup_{g \in \mathcal{G}} \frac{1}{m} \sum_{i=1}^m \sigma_i g(z_i) \right],$$

where independent  $\sigma_i$  are uniform on  $\{-1, +1\}$  and independent of  $S$ . Fix  $S$ , draw  $\sigma$ , maximize separately for that sign vector, then average.

The distribution-averaged version is

$$\mathfrak{R}_{m(\mathcal{G})} = E_{S \sim D^m} [\hat{\mathfrak{R}}_{S(\mathcal{G})}].$$

$\mathcal{H}$  contains predictors;  $\mathcal{G}$  contains their loss functions.

## 2. Ghost sample and symmetrization

A **ghost sample**  $S' = (z_1', \dots, z_{m'}') \sim D^m$  is independent of  $S$  and is used only inside the proof. It is not validation data or extra data given to the learner. With  $\hat{E}_{S[g]} = m^{-1} \sum_i g(z_i)$ ,

$$\begin{aligned} E_S \sup_{g(E_{D[g]} - \hat{E}_{S[g]})} & \\ & \leq E_{S, S'} \sup_{g(\hat{E}_{S'[g]} - \hat{E}_{S[g]})} \\ & = E_{S, S', \sigma} \sup_g \frac{1}{m} \sum_i \sigma_i (g(z_{i'}) - g(z_i)) \\ & \leq 2\mathfrak{R}_{m(g)}. \end{aligned}$$

Factor ledger: 1, 1, 2. The first step uses  $\sup_g E[X_g] \leq E[\sup_g X_g]$ ; exchangeable pairs license the signed equality; only supremum subadditivity creates the factor two.

## 3. Rademacher generalization bounds

For measurable  $\mathcal{G}$  mapping  $\mathcal{Z}$  to  $[0, 1]$ , with probability at least  $1 - \delta$ , simultaneously for every  $g \in \mathcal{G}$ ,

$$E_{D[g]} \leq \hat{E}_{S[g]} + 2\mathfrak{R}_{m(g)} + \sqrt{\frac{\ln(\frac{1}{\delta})}{2m}}.$$

A valid data-dependent form is

$$E_{D[g]} \leq \hat{E}_{S[g]} + 2\hat{\mathfrak{R}}_{S(g)} + 3\sqrt{\frac{\ln(\frac{2}{\delta})}{2m}}.$$

Apply the argument to  $-\mathcal{G}$  or split the confidence budget to control absolute deviations. Constants depend on the one- or two-sided form. The ghost sample makes a paired empirical process; it is not needed because the unknown numerical value of  $R(h)$  is somehow illegal in a probability bound.

## 4. Finite VC closes the arrow

For binary  $\mathcal{H}$  written in  $\{-1, +1\}$  values, let  $V_S = \{(h(x_1), \dots, h(x_m)) : h \in \mathcal{H}\}$ . Massart's lemma gives

$$\hat{\mathfrak{R}}_{S_X}(\mathcal{H}) \leq \sqrt{2 \ln \frac{|V_S|}{m}} \leq \sqrt{2 \ln \Pi_{\mathcal{H}} \frac{m}{m}}.$$

If  $d = \text{VCdim}(\mathcal{H}) < \infty$  and  $m \geq d$ , Sauer-Shelah gives

$$\Pi_{\mathcal{H}}(m) \leq \sum_{j=0}^d \binom{m}{j} \leq \left(e \frac{m}{d}\right)^d.$$

Therefore, with probability at least  $1 - \delta$ ,

$$\sup_{h \in \mathcal{H}} |R(h) - \hat{R}_{S(h)}| = O\left(\sqrt{\frac{d \ln(e \frac{m}{d}) + \ln(\frac{1}{\delta})}{m}}\right).$$

Causal order: Sauer count  $\rightarrow$  Massart random-sign bound  $\rightarrow$  symmetrization/concentration  $\rightarrow$  uniform convergence.

## 4. Rates and model choice

### 1. VC sample-complexity regimes

For finite VC dimension  $d$ :

- **Realizable PAC**: lower order  $\Omega\left(\frac{d + \ln(\frac{1}{\delta})}{\epsilon}\right)$ ; a standard upper bound is  $O\left(\frac{d \ln(\frac{1}{\epsilon}) + \ln(\frac{1}{\delta})}{\epsilon}\right)$ .
- **Agnostic PAC / uniform convergence**:  $\Theta\left(\frac{d + \ln(\frac{1}{\delta})}{\epsilon^2}\right)$  up to absolute constants.

Halving  $\epsilon$  costs roughly  $2 \times$  samples in the realizable regime (plus logs), but  $4 \times$  in the agnostic regime. Realizability permits consistency detection; noisy agnostic learning must estimate differences between nonzero risks.

## 2. Structural risk minimization (SRM)

Let  $\mathcal{H} = \cup_{k \in \mathbb{N}} \mathcal{H}_k$ . Give every active class a weight  $w_k > 0$  with  $\sum_k w_k = 1$ , set  $\delta_k = w_k \delta$ , and use a justified uniform radius  $\varepsilon_{k(m, \delta_k)}$ . Then, with probability at least  $1 - \delta$ , all active class certificates hold simultaneously:

$$R(h) \leq \hat{R}_{S(h)} + \varepsilon_{k(m, w_k \delta)}, \quad h \in \mathcal{H}_k.$$

SRM minimizes the right side across both  $h$  and  $k$ . A zero-weight class is excluded (equivalently, has infinite penalty). If a hypothesis lies in several classes, use its smallest valid assigned bound.

The family, confidence allocation, and theorem supporting each penalty must be declared; an attractive empirical-risk-plus-square-root expression alone is not the stated distribution-free certificate.

## 3. Select the proof route

- Fixed finite class, uniform upper bound: Hoeffding + union bound; decisive quantity  $\ln|\mathcal{H}|$ .
- Binary finite-VC class: Sauer/growth + Massart; decisive quantities  $\Pi_{\mathcal{H}}(m)$  and  $d$ .
- Real-valued loss or useful sample geometry: ghost sample + Rademacher symmetrization; decisive quantity  $\mathfrak{R}_{S(g)}$ .
- Overstrong universal learner/rate claim: shattered-set indistinguishable labels for a lower bound.

Use an upper-bound method for a generalization certificate and a shattered-set construction for impossibility. Infinite cardinality alone does not choose the route.